Dietary total antioxidant capacity and gastric cancer risk in the European prospective investigation into cancer and nutrition study.

A high intake of dietary antioxidant compounds has been hypothesized to be an appropriate strategy to reduce gastric cancer (GC) development. We investigated the effect of dietary total antioxidant capacity (TAC) in relation to GC in the European Prospective Investigation into Cancer (EPIC) study including 23 centers in 10 European countries. A total of 521,457 subjects (153,447 men) aged mostly 35-70 years old, were recruited largely between 1992 and 1998. Ferric reducing antioxidant potential (FRAP) and total radical-trapping antioxidant parameter (TRAP), measuring reducing and chain-breaking antioxidant capacity were used to measure dietary TAC from plant foods. Dietary antioxidant intake is associated with a reduction in the risk of GC for both FRAP (adjusted HR 0.66; 95%CI (0.46-0.95) and TRAP (adjusted HR 0.61; 95%CI (0.43-0.87) (highest vs. lowest quintile). The association was observed for both cardia and noncardia cancers. A clear effect was observed in smokers with a significant reduction in GC risk for the fifth quintile of intake for both assays (highest vs. lowest quintile: adjusted HR 0.41; 95%CI (0.22-0.76) p for trend <0.001 for FRAP; adjusted HR 0.52; 95%CI (0.28-0.97) p for trend <0.001 for TRAP) but not in nonsmokers. In former smokers, the association with FRAP intake was statistically significant (highest vs. lowest quintile: adjusted HR 0.4; 95%CI (0.21-0.75) p < 0.05); no association was observed for TRAP. Dietary antioxidant capacity intake from different sources of plant foods is associated with a reduction in the risk of GC.